



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 29/06/2017*

EXERCISE

It is given a Moving Average (MA) process $S[n]$ of order 1, $S[n] = I[n] - 0.8 \cdot I[n-1]$, where $I[n]$ is the zero-mean innovation process with variance $\sigma_I^2 = 2$.

(1) Derive the system function $L(z)$ of the innovation filter, its zeros and poles, and determine if it is a low-pass, high-pass, or band-pass filter. Derive the autocorrelation function (ACF) $R_s[m] = E\{S[n]S[n+m]\}$ and the power spectral density (PSD) $S_s(e^{j2\pi f}) = FT\{R_s[m]\}$ of the process $S[n]$ and plot them.

(2) Derive the optimal linear predictor of order $P=2$ of the process $S[n]$, i.e. the linear minimum mean square error (LMMSE) estimator of $S[n]$ given the observed data $S[n-1]$ and $S[n-2]$; derive its mean square error (MSE) $\sigma_\varepsilon^2 \triangleq E\{\varepsilon^2[n]\} = E\{(S[n] - \hat{S}[n])^2\}$.

(3) Assuming that the signal of interest $S[n]$ is observed in the presence of additive white noise, i.e. $X[n] = S[n] + W[n]$, where $W[n]$ is additive white noise with variance $\sigma_w^2 = 1$, derive the optimal linear predictor of order $P=2$ of the process $S[n]$, i.e. the LMMSE estimator of $S[n]$ given the observed data $X[n-1]$ and $X[n-2]$; derive its MSE and compare it with the one previously derived.



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

Solution.

$$(1) S[n] = I[n] - 0.8I[n-1]$$

$$S(z) = I(z) - 0.8z^{-1}I(z)$$

$$L(z) = \frac{S(z)}{I(z)} = 1 - 0.8z^{-1} = \frac{z - 0.8}{z}$$

The innovation filter has 1 zero at $z = 0.8$ and 1 pole at $z = 0$. Hence, the innovation filter is a high-pass filter, and so the process is a high-pass process.

The ACF and the PSD are given by:

$$\begin{aligned} R_s[m] &= E\{S[n]S[n+m]\} \\ &= E\{(I[n] - 0.8 \cdot I[n-1])(I[n+m] - 0.8 \cdot I[n+m-1])\} \\ &= R_I[m] - 0.8 \cdot R_I[m-1] - 0.8 \cdot R_I[m+1] + 0.64 \cdot R_I[m] \\ &= 1.64 \cdot \sigma_I^2 \delta[m] - 0.8 \cdot \sigma_I^2 \delta[m-1] - 0.8 \cdot \sigma_I^2 \delta[m+1] \\ &= 3.28 \cdot \delta[m] - 1.6 \cdot \delta[m-1] - 1.6 \cdot \delta[m+1] \end{aligned}$$

$$R_s[m] = \begin{cases} 3.28, & m = 0 \\ -1.6, & m = \pm 1 \\ 0, & |m| \geq 2 \end{cases}$$

$$\begin{aligned} S_s(e^{j2\pi f}) &= FT\{R_s[m]\} = \sum_{m=-\infty}^{+\infty} R_s[m]e^{-j2\pi fm} \\ &= \sum_{m=-\infty}^{+\infty} [3.28 \cdot \delta[m] - 1.6 \cdot \delta[m-1] - 1.6 \cdot \delta[m+1]]e^{-j2\pi fm} \\ &= 3.28 - 1.6e^{-j2\pi f} - 1.6e^{j2\pi f} \\ &= 3.28 - 3.2\cos(2\pi f) \end{aligned}$$

$S[n]$ is a high-pass process.



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

$$(2) \quad \hat{S}[n] = h[1]S[n-1] + h[2]S[n-2] + b$$

Since all the processes are zero mean we have that: $E\{\varepsilon[n]\} = 0 \Rightarrow b = 0$.

$$E\{\varepsilon[n]S[n-m]\} = 0, \quad m=1,2$$

$$E\{(S[n] - \hat{S}[n])S[n-m]\} = 0, \quad m=1,2$$

$$E\{(S[n] - h[1]S[n-1] - h[2]S[n-2])S[n-m]\} = 0, \quad m=1,2$$

$$R_s[m] - h[1]R_s[m-1] - h[2]R_s[m-2] = 0, \quad m=1,2$$

$$\begin{cases} h[1]R_s[0] + h[2]R_s[-1] = R_s[1] \\ h[1]R_s[1] + h[2]R_s[0] = R_s[2] \end{cases}$$

$$\begin{bmatrix} R_s[0] & R_s[-1] \\ R_s[1] & R_s[0] \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} R_s[1] \\ R_s[2] \end{bmatrix}$$

$$\begin{bmatrix} 3.28 & -1.6 \\ -1.6 & 3.28 \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} -1.6 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} -0.6401 \\ -0.3123 \end{bmatrix}$$

$$\begin{aligned} \sigma_\varepsilon^2 &= E\{\varepsilon^2[n]\} = E\{(S[n] - \hat{S}[n])S[n]\} \\ &= E\{(S[n] - h[1]S[n-1] - h[2]S[n-2])S[n]\} \\ &= R_s[0] - h[1]R_s[1] - h[2]R_s[2] \\ &= 3.28 + 0.6401 \cdot (-1.6) + 0.3123 \cdot 0 \\ &= 2.2558 \end{aligned}$$



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

$$(3) \quad \hat{S}[n] = h[1]X[n-1] + h[2]X[n-2] + b$$

Since all the processes are zero mean we have that $E\{\varepsilon[n]\} = 0 \Rightarrow b = 0$.

$$E\{\varepsilon[n]X[n-m]\} = 0, \quad m = 1, 2$$

$$E\{(S[n] - \hat{S}[n])X[n-m]\} = 0, \quad m = 1, 2$$

$$E\{(S[n] - h[1]X[n-1] - h[2]X[n-2])X[n-m]\} = 0, \quad m = 1, 2$$

$$R_{SX}[-m] - h[1]R_X[m-1] - h[2]R_X[m-2] = 0, \quad m = 1, 2$$

$$\begin{cases} h[1]R_X[0] + h[2]R_X[-1] = R_{SX}[-1] = R_S[1] \\ h[1]R_X[1] + h[2]R_X[0] = R_{SX}[-2] = R_S[2] \end{cases}$$

$$\begin{bmatrix} R_X[0] & R_X[-1] \\ R_X[1] & R_X[0] \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} R_S[1] \\ R_S[2] \end{bmatrix}$$

$$R_X[m] = R_S[m] + R_W[m] = R_S[m] + \sigma_w^2 \delta[m] = R_S[m] + \delta[m]$$

$$\begin{bmatrix} 4.28 & -1.6 \\ -1.6 & 4.28 \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} -1.6 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} -0.4346 \\ -0.1625 \end{bmatrix}$$

$$\begin{aligned} \sigma_\varepsilon^2 &= E\{\varepsilon^2[n]\} = E\{(S[n] - \hat{S}[n])S[n]\} \\ &= E\{(S[n] - h[1]X[n-1] - h[2]X[n-2])S[n]\} \\ &= R_S[0] - h[1]R_S[1] - h[2]R_S[2] \\ &= 3.28 + 0.4346 \cdot (-1.6) + 0.1625 \cdot 0 \\ &= 2.5846 \end{aligned}$$

which is obviously greater than the MSE in the previous case, because of the presence of additive noise $W[n]$.